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# Radar2ECG: Multi-Scale Bottleneck Fusion and Cross-modal Semantic Distillation for Conditional Electrocardiogram Generation from Radar Heart Sound

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***Abstract*—The field of conditional Electrocardiogram(ECG) generation focuses on generating specified ECGs under given conditions for medical purposes. Existing methods are typically based on conditions of simple inputs like text or lead types. However, they struggle to handle the complexity of radar heart sound signals due to the lack of effective feature extraction, which hinders capturing the intricate waveform correlations between radar heart sounds and ECGs. Considering that radar-detected heart sound signals are contactless, the application is of essential value in a real-world deployment like sleep scenarios. Moreover, no prior approaches have addressed this specific task. To tackle this challenge, we propose a novel multi-scale feature fusion network framework, Radar2ECG. This model leverages pre-trained autoencoders for heart sound and ECG signals, aligning and integrating multi-layer features through a bottleneck structure to enhance receptive fields and reduce redundant features, thereby capturing the correlations between heart sounds and ECGs. Finally, we employ knowledge distillation to transfer knowledge from the ECG decoder to the heart sound decoder. We present three anomaly type datasets and extensive experiments conducted on both normal and abnormal datasets demonstrate that our method outperforms existing models in both accuracy and robustness. The multi-scale feature fusion significantly improves performance, showcasing strong potential in ECG generation and heart sound anomaly detection tasks.**

***Index Terms*—Radar-to-ECG, ECG synthesis, knowledge distillation**

## I. Introduction

Heart sounds (HS) and electrocardiograms (ECG) provide crucial insights into cardiac function and potential pathological conditions, and are widely regarded as fundamental tools for diagnosing coronary artery disease, heart failure, and arrhythmias [1], [2]. In clinical settings, heart sounds are typically measured using a stethoscope, while ECG signals are recorded through adhesive electrodes placed on the skin. **These contact-based measurement techniques impose strict environmental and procedural requirements, indirectly limiting the availability of large-scale heart sound and ECG data.** In recent years, non-contact sensing has become a focal point, especially for long-term continuous monitoring during sleep. Radar technology has shown promise in non-contact heart sound acquisition [3]. However, due to the unique nature of ECG signal generation, non-contact methods for direct ECG acquisition remain in the early stages of development.

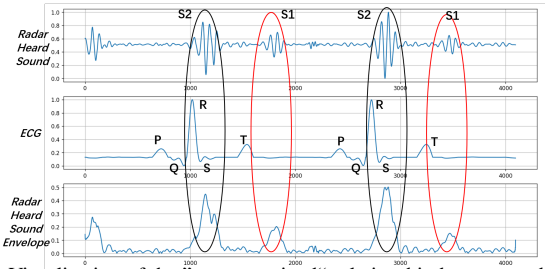



Fig. 1. Visualization of the "coarse-grained" relationship between radar heart sound and synchronized ECG signals shows that the S1 wave of the heart sound correlates with the onset of the P wave after the ECG's T wave, while the S2 wave correlates with the onset of the S wave after the R wave.

In order to generate ECG signals, the vast majority of methods build upon one-dimensional time series generative models, making various improvements. These range from autoencoders and their variants [4]–[6] to Generative Adversarial Networks (GANs) that incorporate discriminators and combine with networks like Recurrent Neural Networks (RNNs) and transformers [7]–[10], as well as the more recent diffusion probabilistic models [11]–[13]. However, most of these approaches are used for unconditional generation, aiming to produce highly diverse and high-fidelity ECGs. In the case of more difficult conditional generation, where ECG

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signals are generated based on the correlation between heart sounds and ECGs as illustrated in Fig.1, the results remain unsatisfactory. The inherent complexity of heart sound signals and the uncertainty in aligning them with ECG signals make it more challenging to generate stable ECGs compared to other conditional generation tasks. To our knowledge, no prior works have investigated this problem: using radar signals to generate ECG.

**On the one hand**, compared to mainstream ECG generation methods that rely on text or categorical inputs, **the radar heart sounds contain far more information than texts or categories (See Fig.1 for details)** , which are temporally aligned with ECGs. Combined with the scarcity of relevant datasets, effectively extracting this information presents a significant challenge for ECG generation. **On the other hand**, due to **the weak correlation between radar heart sounds and ECG signals (See Fig.1 for details)**, there are inherent difficulties in perfectly aligning the waveforms. Addressing the misalignment between waveforms is another key challenge in generating ECGs conditioned on heart sounds.

To address the limitations of existing methods, we propose a multi-scale feature fusion network based on an encoder-decoder architecture. Instead of using heart sounds as a straightforward embedded feature condition for generating ECG signals, we utilize a complete encoder module, such as a 1D ResNet-50 network, to extract multi-layered features from the radar heart signals. This approach leverages the encoder as the core component for ECG generation. Compared to the encoders used in VAE and GAN frameworks, **we proposed a multi-scale fusion bottleneck module**, that is capable of capturing the complex waveform characteristics inherent in ECGs and explicitly extracting features across different scales, from local to global.

On the decoder side, we employ a network that mirrors the encoder, such as a 1D DeResNet-50, to decode the features. To better align the waveforms and enhance the correlation between heart sounds and ECG signals, we depart from the traditional direct connection between the encoder and decoder. Instead, we introduce a bottleneck structure to align and fuse the multi-scale outputs from the encoder. During the training phase, we applied **multi-layer semantic distillation** to transfer the knowledge of ECG feature generation from the pre-trained ECG decoder to the Radar2ECG decoder. Our experiments demonstrate that the proposed structure effectively eliminates redundant information, focuses on the correspondence of local features, and fuses multi-scale feature sequences to reduce alignment errors between heart sound and ECG signals at the feature level. This leads to more stable and accurate ECG generation results, outperforming other baseline models.

## II. Related Work

Generative models, initially thriving in the field of image processing, have gradually expanded to the domain of time series analysis, with fundamental models such as autoencoders and their variants [4]–[6], as well as Generative Adversarial Networks (GANs) and their derivatives [7]–[10], which produce clearer results. Recently, diffusion probabilistic models [11] have emerged as powerful tools capable of generating higher-fidelity and more diverse 1D time series.

However, time-series generative models still face challenges when applied to ECG signal generation. Firstly, structural limitations of certain models [9], [10], [14] restrict them to generating only short-term ECG signals, making them inadequate for long-term ECG generation. Secondly, the lack of diversity and scarcity in ECG datasets limits the models' ability to generate diverse samples [7], [15]. Lastly, while some methods have achieved conditional ECG generation under specific conditions [4], [5], [9], most remain confined to unconditional generation [7], [9], [10], [14], [16].

## III. Method

### A. Overall

The proposed architecture, as shown in Fig.2, is composed of three key components: an encoder, a decoder, and a multi-scale fusion bottleneck structure. We replace simple embedding encoders with a full encoder to better capture the complexity of heart sound feature extraction. By aligning and fusing multi-layered features, we compact the feature representation, effectively removing redundancy and strengthening the correlation between heart sounds and ECG signals. The training process is divided into two stages: *(a) the pre-training phase and (b) the distillation phase*.

Let $P = \{(s^i_{hs}, s^i_{ecg})\}^n_{i=1}$ represent the training dataset consisting of paired heart sound signals and ECG signals, where $s^i_{hs}$ represents the heart sound signal in the $i$-th signal pair, $s^i_{ecg}$ represents the corresponding ECG signal in the same pair and $n$ denotes the total number of signal pairs. **In the pre-training phase**, autoencoders for both heart sound and ECG signals are trained to ensure that the self-generated signal $\hat{S}_{ecg}$ closely resembles the original signal $s \in P$ , effectively preparing the encoder and decoder to capture and store a rich representation of features for the subsequent training phase. **In the distillation phase**, the knowledge related to ECG feature reconstruction from the ECG decoder is distilled into the heart sound-to-ECG decoder. The bottleneck structure is adapted, enabling the generation of ECG signals from heart sounds, facilitating an effective transformation from heart sound to ECG.

### B. Multi-scale Fusion Architecture

To construct a deeper network capable of extracting rich features from the complex and variable heart sound signals, we employ the first three layers of a one-dimensional ResNet network as the encoder, while the decoder consists of all layers of a one-dimensional DeResNet network, which mirrors the encoder.

Unlike other encoder-decoder architectures, such as autoencoder(AE) [4]–[6] and generative adversarial networks (GANs) [7]–[10], where the decoder directly receives the output from the final layer of the encoder, relying on deep neural layers to extract global abstract features for reconstruction,

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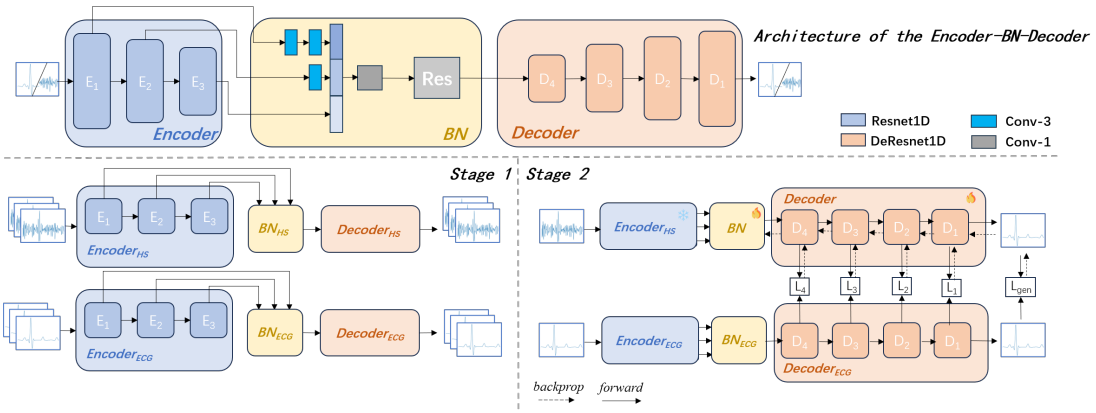



Fig. 2. Framework Overview: The upper section shows the encoder-bottleneck-decoder architecture, where the bottleneck aligns multi-scale features from different encoder layers to compact features, remove redundancy, and improve decoding. The lower section outlines two training phases: pre-training, where the model generates its own signals, and distillation, where the Radar2ECG decoder learns ECG generation from the ECG decoder, with adjustments to the bottleneck to support feature transformation.

these approaches often overlook the local semantic information captured by shallow layers. This omission is detrimental to understanding the misalignment between heart sound signals and ECG signals.

To address this limitation, we propose a multi-scale fusion module as a bottleneck structure that connects the output of the encoder to the input of the decoder, as shown in layer BN in Fig.2. We apply one or more one-dimensional convolutional layers to downsample the shallow features and achieve feature alignment, connecting feature maps from different layers. A one-dimensional convolution layer with a kernel size of 1 is used for feature condensation, removing redundant information to emphasize the correlation between heart sound and ECG signals. Finally, this bottleneck structure connects to the fourth layer of the one-dimensional ResNet network, serving as the input to the decoder. Compared to U-Net, which employs skip connections between the encoder and decoder at each level, our method of feature alignment and fusion effectively corrects feature misalignment before the features reaches the decoder. This prevents the propagation of misaligned features into the decoder, thereby reducing generation errors.

### C. Inter-decoder knowledge distillation

To achieve feature alignment on the generative side, we employ knowledge distillation by using the entire ECG autoencoder as the teacher model. The encoder from the heart sound autoencoder, along with a newly designed bottleneck structure and decoder, is used as the student model.

Given a signal pair $(s_{hs}, s_{ecg}) \in \{(s^i_{hs}, s^i_{ecg})\}^n_{i=1}$ , their respective projections in the bottleneck structures of the student and teacher models are represented as $\phi_s, \phi_t$. The pairing relationship of the signal pair in the corresponding decoders is represented as $\{f^k_s = D^k_s(\phi_s), f^k_t = D^k_t(\phi_t)\}$, where $D^k_s$ and $D^k_t$ are the decoders at the $k$-th layer in the student and teacher models, respectively. $f^k_s, f^k_t \in \mathbb{S}^{C^k \times L^k}$, where $C^k$ and $L^k$ represent the number of channels and the sequence length of the activation features at the $k$-th layer, respectively.

We compute the cosine similarity along the channel dimension between the corresponding feature sequences to more accurately capture the relationship between high-dimensional and low-dimensional information. The similarity across different layers is then summed to obtain the feature loss. The formula is as follows:

$$L_{\text{feat}} = \sum_{k=1}^{K} \left\{ \frac{1}{L^k} \sum_{l=1}^{L} 1 - \frac{f^k_s(l) \cdot f^k_t(l)}{\|f^k_s(l)\| \|f^k_t(l)\|} \right\} \tag{1}$$

Finally, the output of the first layer of the student decoder, which is the ECG signal $\hat{S}_{ecg}$ generated from the heart sound, is compared with the corresponding real ECG signal using mean squared error to obtain the generation loss. This is expressed as follows:

$$L_{\text{gen}} = \frac{1}{n} \sum_{i=1}^{n} \left( S^i_{ecg} - \hat{S}^i_{ecg} \right)^2 \tag{2}$$

The two losses are summed to obtain the final loss function:

$$L_{\text{final}} = \lambda_1 L_{\text{feat}} + \lambda_2 L_{\text{gen}} \tag{3}$$

Where $\lambda_1$ and $\lambda_2$ are hyperparameters that balance the contribution of the two losses.

## IV. Experiments and Results

### A. Experiment Setup

**Benchmarks and Datasets:** We validate our approach using two datasets: the Scidata Phase dataset [3], containing only normal samples, to evaluate the effectiveness of generating ECG signals from radar-based heart sounds, and three synthetic abnormal datasets to assess the potential of multi-scale feature fusion for detecting anomalies in radar heart sounds.

The Scidata Phase dataset includes radar-based vital signs and synchronized reference signals from thirty healthy individuals across five conditions. Since the resting state data was available for all individuals, we used 19,048.6 seconds of data from the resting condition, extracting radar heart sounds and ECG signals. After filtering, denoising, and normalization, we applied a sliding window (width: 4096, step: 1024) for data

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augmentation. The dataset was split 8:2 into 29,680 training and 7,420 testing samples.

Since the Scidata Phase dataset contains only normal samples, we generated three types of corresponding abnormal samples for each normal instance using the following methods: (1) cardiac arrest: setting random segments of the sample to 0.5, (2) environmental interference: adding noise, and (3) rapid heart rate: mixing samples to create hybrid signals, are shown in (b)(c)(d) of Fig.3, respectively.

**Evaluation Metrics:** The evaluation metrics are categorized into two types: distance-based (RMSE, MAE) where lower values indicate better performance, and correlation-based (cosine similarity, $R^2$score) where higher values signify better performance.

**Comparision Methods:** We categorized the comparison methods into two groups: generative models, such as CVAE [4], CWGAN [4], and SSSD [11], and sequence-to-sequence models, including LSTM [17] and Unet [18].

**Implementations:** In the experiments, the encoder utilizes a 1D ResNet50 network and the decoder utilizes a 1D DeResNet50 network. Our model was trained with a batch size of 32, over 300 epochs, with a learning rate of 0.001. We employed the Adam optimizer with $\beta = (0.9, 0.999)$.

TABLE I
COMPARISON OF RESULTS WITH OTHER GENERATED ECG MODELS; RELATIVE GAIN IS RELATED TO THE SECOND BEST METHOD.

| | RMSE↓ | MAE↓ | COS↑ | $R^2$score↑ |
|---|---|---|---|---|
| CVAE [4] | 0.155 | 0.101 | 0.825 | -0.798 |
| CWGAN [4] | 0.990 | 0.788 | -0.0231 | -0.650 |
| LSTM [17] | 0.153 | 0.102 | 0.824 | -0.664 |
| Unet [18] | 0.155 | 0.0908 | 0.831 | -0.818 |
| SSSD [11] | 0.205 | 0.130 | 0.627 | -2.083 |
| Ours | **0.0237** | **0.0134** | **0.987** | **0.879** |
| Relative gain | 84.5% | 85.2% | 18.8% | 142% |

TABLE II
RESULTS OF FUSION OF DIFFERENT FEATURE LAYERS

| | RMSE↓ | MAE↓ | COS↑ | $R^2$score↑ |
|---|---|---|---|---|
| *Config.A* | 0.152 | 0.119 | 0.861 | -0.568 |
| *Config.B* | 0.0340 | 0.0236 | 0.984 | 0.847 |
| *Config.C* | 0.0294 | 0.0188 | 0.984 | 0.859 |
| *Config.D* | 0.0357 | 0.0235 | 0.981 | 0.830 |
| *Config.E* | **0.0237** | **0.0134** | **0.987** | **0.879** |

### B. Results

We trained and tested both the comparison methods and our model on the Scidata Phase dataset, as shown in Table I. Our method consistently outperforms existing approaches, with improvements ranging from 18.8% to 142% across different metrics. This performance boost is attributed to the multi-scale feature fusion, which effectively captures the alignment between heart sound and ECG signals.

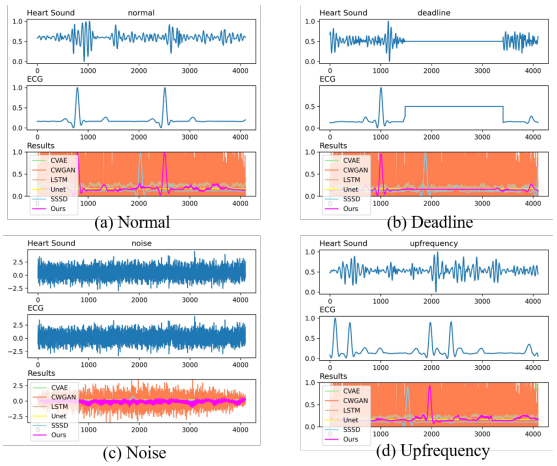
### C. Analysis

**Ablation Study:** To validate the effectiveness of bottleneck fusion, we experimented with various combinations of the three encoder feature maps. Configurations included: the first and third layers (*Config.A*), second and third layers (*Config.B*), first and second layers (*Config.C*), third layer alone (*Config.D*), and all three layers (*Config.E*). As shown in Table II, the three-layer fusion (*Config.E*) outperformed the others.

For anomaly detection, we tested on synthetic abnormal datasets, splitting them 8:2 for training and testing. Identical classifiers were added to the encoder and bottleneck, and these classifiers were applied to other comparative methods as well, with results in Table III showing significant improvements in detecting abnormal heart sounds with feature fusion.

TABLE III
ACCURACY OF DIFFERENT METHODS AND SETTINGS UNDER THE ANOMALY DETECTION TASK

| | Add DeadLine | Add Noise | Mixing Frequency | Avg. |
|---|---|---|---|---|
| CVAE | 79.0% | 100% | 75.9% | 84.9% |
| Unet | 91.5% | 100% | 79.7% | 90.4% |
| CWGAN | 79.9% | 80.0% | 79.8% | 79.9% |
| LSTM | 80.8% | 79.7% | 79.8% | 80.1% |
| Ours without feature fusion | 86.6% | **100%** | 98.6% | 95.1% |
| Ours | **98.0%** | **100%** | **99.1%** | **99.0%** |



(a) Normal (b) Deadline (c) Noise (d) Upfrequency

Fig. 3. Visualisation of ECGs generated by different methods under different kinds of datasets.

**Qualitative Analysis:** We trained our proposed method and the baseline methods from Section IV-A on the normal dataset and tested on both normal and three types of synthetic abnormal datasets. The ECG visualizations generated from the test set heart sounds (Fig.3) demonstrate that, compared to the baselines, our model exhibits strong generalization. Whether generating ECG from in-distribution (normal) or out-of-distribution (abnormal) heart sounds, our method consistently produces waveforms closely resembling the ground truth, showcasing its potential for anomaly detection tasks.

## V. CONCLUSION

In this work, we propose a novel radar heart sound-to-ECG generation task and introduce a conditional generative model. Our method is based on an encoder-decoder architecture, where the multi-scale fusion of multi-layer features from the encoder replaces the final layer output as input to the decoder. We pretrain autoencoders for both heart sound and ECG signals, and apply semantic distillation from the ECG decoder to the radar heart sound decoder. Through comprehensive evaluation, our model, Radar2ECG, demonstrates superior performance in radar heart sound-to-ECG generation compared to baseline models, while also showing significant potential in heart sound anomaly detection.

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